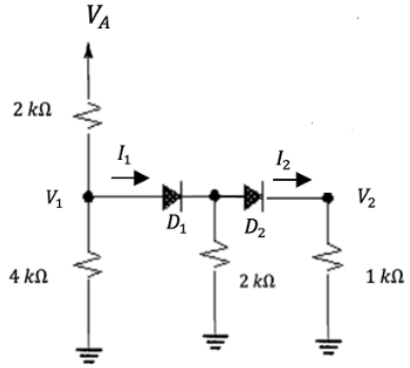
Quiz (1) Solution

1) Find the values of I_1, I_2, V_1 and V_2 in the circuit below. Use the ideal diode model. Show all steps. Put answers in table.

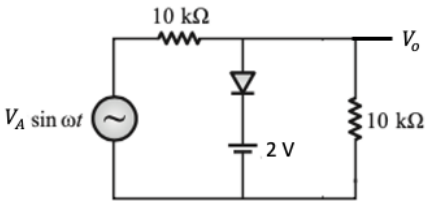


Both diodes are ON

$$V_1 = V_2 = V_A \frac{4\,||2||1}{2 + 4\,||2||1}$$

$$I_2 = \frac{V_2}{1k} \qquad I_1 = \frac{V_1}{2k} + I_2$$

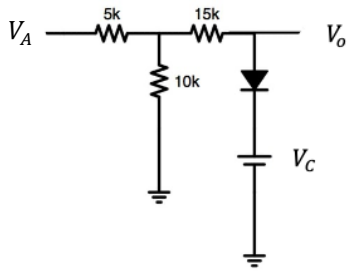
2) Use the ideal diode model in the circuit below and find $V_{o(\max)}$ and $V_{o(\min)}$. Show all steps.



$V_A < 4\,V$ *diode OFF* $\quad V_{o(\max)} = \frac{V_A}{2} \quad V_{o(\min)} = -V_A/2$

$V_A > 4\,V$ *diode ON* $\quad V_{o(\max)} = 2\,V \quad V_{o(\min)} = -V_A/2$

3) For the circuit below, find the output V_o for each value of V_C and record in the table. Take $V_{DO} = 0.7\,V$ for the diode. Show all steps.



Diode ON for $\frac{10}{15}V_A > V_C + V_{DO} \rightarrow V_o = V_C + V_{DO}$

Diode OFF for $\frac{10}{15}V_A < V_C + V_{DO} \rightarrow V_o = \frac{10}{15}V_A$